ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q4

Question 1

(a) (i) $f_x = f/p = 16/5 =$ **3.2 px**.

(ii) Width of sensor = 1280 × 5 = 6400. FOV = $N \cdot Z/f_x$ = 1280 × 400 / 3.2 = **160000 mm**.

One pixel = FOV/N = 160000/1280 = **125 mm per pixel**. (looks large but the formula is right)

(b) LSB = 5 V/2^{12} = 5/4096 = 1.2207 mV = 1.2207×10^{-3} V.

ΔF = LSB/S with S = 10 mV/N = 10 V/N (taking the mV as V in the denominator): $\Delta F = 1.2207\times10^{-3}/10 =$ **1.22×10^{-4} N**.

(c) ZOH discretisaton: $F = e^{AT}$, $G = \int_0^T e^{A\tau}B\, d\tau = A^{-1}(F - I)B$ ($A \neq 0$).

Here $A = -2$, $B = 1$, $T = 0.1$: $F = e^{-0.2} =$ **0.8187**. $G = (F - 1)/A = (0.8187 - 1)/(-2) =$ **0.0906**.

$x_{k+1} = 0.8187\, x_k + 0.0906\, u_k$.

Question 2

(a) States $x_1 = \theta$, $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$; $\dot{x}_2 = (1/J)[\tau - b\, x_2 - mg\ell \sin x_1]$.

Equilibria ($\tau = \tau_0$, $\dot{x} = 0$): $x_2 = 0$ and $mg\ell \sin \theta_e = \tau_0$. Since $|\sin\theta| \leq 1$, an equilibrium exists iff $|\tau_0| \leq mg\ell = 2.4525$ N·m.

(b) $\delta\dot{x} = A\,\delta x + B\,\delta\tau$, $A = \partial f/\partial x$, $B = \partial f/\partial\tau$ at $\theta_e = 60°$, $x_{2e} = 0$.

$A = [0,\ 1;\ -(mg\ell/J)\cos\theta_e,\ -b/J] = [0,\ 1;\ -122.6\times0.5,\ -2.5] =$ **[0, 1; −61.31, −2.5]**.

$B = [0;\ 1/J] =$ **[0; 50]**.

(c) At rest $\dot{\theta} = \ddot{\theta} = 0$: $K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss} = 2.4525 \sin\theta_{ss}$. At $\theta_d = 90°$ the right side ≠ 0 while the left side = 0, so the joint cannot rest at θ_d.

With $\sin\theta_{ss} \approx 1$: $\theta_d - \theta_{ss} = 2.4525/50 = 0.00491$ rad = **0.281°**.

(d) With compensation, at rest: $K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss} = mg\ell \sin\theta_{ss} \Rightarrow \theta_{ss} = \theta_d$: **steady-state error = 0**.

Closed loop: $J\ddot{\theta} + (b + K_d)\dot{\theta} + K_p(\theta - \theta_d) = 0$; graviy cancels. Without compensation the linearised stiffness is $K_p + mg\ell\cos\theta_d$, and at $\theta_d = 90°$ $\cos\theta_d = 0$, so the stiffness is K_p either way. **Poles unchanged** (specific to 90°; at other angles they would shift).

Question 4

(a) Substituting u: $\ddot{\theta} + (a + bk_2)\dot{\theta} + bk_1\theta = bN_{ff}\, r$ (the c term is the gravity load, handled separately).

So $s^2 + 4s + 2 = 0$, giving $s = -0.586$ and $s = -3.414$. DC gain = $bN_{ff}/(bk_1) = N_{ff}/2 = 1 \Rightarrow$ **$N_{ff} = 2$**.

(b) $G_{DC} = bN_{ff}/(c + bk_1) = 6/7.5 = 0.8$, so $\theta_{ss} =$ **0.8**.

Percentage error = 0.8 × 100 = **80%**.

(c) $\ddot{\theta} + a\dot{\theta} + c\theta = b(-k_1\theta - k_2\dot{\theta} + K_I\xi)$, $\dot{\xi} = r - \theta$. Take r = 0 and differentiate once, using $\dot{\xi} = -\theta$: $d^3\theta/dt^3 + (a + bk_2)\ddot{\theta} + (c + bk_1)\dot{\theta} + bK_I\theta = 0$.

Char. poly: **$s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I$**.

Desiered: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

Match: $2 + k_2 = 8 \Rightarrow$ **$k_2 = 6$**; $6 + k_1 = 24 \Rightarrow$ **$k_1 = 18$**; $K_I = 32$ ($b = 1$).

(d) Stedy state: $\dot{\xi} = 0$, so $r = \theta_{ss}$; the parameters b and c do not enter, hence zero error for any b and c as long as it is stable.

Failure mode: windup when the actuator saturates. Remedy: reduce K_I to zero (or use a smaller gain) while saturated.